

Student Research Plan

PROJECT INFORMATION

Student Name(s): Gnaneswar Kumar Pullaganti , Dakshith Rajkumar , and Dakshith Karthikeyan

Student Grade: All of us are in 7th grade

Student School: Pat and Catherine Fowler Middle School

Title of Project: P.E.N.S.I.L.: Planetary Expeditionary Navigator and Settlement Infrastructure Launcher

Category (select from this list) only list the main header, not bulleted subtopics): Engineering

Technology: Statics and Dynamics (ETSD)

(ETSD)

Team Project (yes or no): Yes

Date you plan to start (if research plan is approved): Once research plan has been approved, we will start our data extraction(e.g. 10/20-10/25).

ALWAYS USE COMPLETE SENTENCES AND ANSWER THE FOLLOWING

Research Location: *(Where do you plan to do your research check all that apply)*

- ☒ Home
- ☐ School
- ☐ Regulated Research Institution (i.e. University Lab etc.)
- ☐ Place of Business, Commercial Property or Industrial Setting
- ☐ Field or Other Public Location (i.e. City Park, Lake, River, Shopping Mall etc.)
- ☐ Other (Neighbor's house, mentor's house etc.)

Location Address: *(For each location above, list the name and give the street address if available, or if not give location description or GPS coordinates)*

9300 Coit Rd Apt# 310, 2013, 2026, Plano, TX 75025

Rationale: *(Why is this project important? Include a brief summary of the background that supports your research problem and explain why this research is important scientifically and if applicable, explain any societal impact of your research)*

This project has deep roots within distant planet exploration. Traditional rovers go at speeds of “1 centimeter per second” (NASA 6). Our goal for this project is to speed up planet exploration by hundreds or thousands of times. The societal impact of our project is to provide astro-scientists a faster, more efficient method for space exploration by using the multiplicity of the rover swarm to our advantage.

Research Question, Hypothesis/Engineering Goal: *(State your RESEARCH QUESTION(S), HYPOTHESIS/ENGINEERING GOAL(S), EXPECTED OUTCOMES. How is this related to your rationale for choosing this project? Traditional projects will have a hypothesis. Engineering projects will have an engineering goal.)*

Can we design a rover system (possibly with multiple programmed swarms of rovers) that is faster than conventional rovers in exploring a planet and more reliable in gathering dependable data and information about the planet?

Materials: *(List the materials you will be using, including chemicals and amounts and concentrations. Include where you purchased any bacteria or other potentially hazardous biological agents. If you used E. coli k-12, make sure to list that as the material. Be extremely detailed with this list. Include the amounts of each material, too. If you are unsure, make your best attempt. This list can always be updated once you get to the final presentation board.)*

- 3 Orange Pi 5B (4GB RAM)
- 3 Yahboom Silan LiDAR
- 3 MPU-6050 Module (Gyroscope + Accelerometer)
- 3 Raspberry Pi Camera Module
- 6 PCA9685 16-Channel Servo Driver
- 3 sets of 40 Male-to-female Jumper Wires
- 3 Portable USB Power Banks
- 3 Small Project Box and Mounting Hardware
- 3 MicroSD Card (32GB or 64GB)
- 3 Electric RC Car (with Alkaline batteries, ESCs, AC motors, and steering servos)
- Computer for JavaScript and Python Programming
- 1 Soldering Iron
- 30 Header Pins
- 30 Solder
- Hot Glue Gun
- Database (e.g., MongoDB installed on a shared computer)
- Wi-Fi Network

Procedure: *(This is the section that will cause most students to have their project returned by the teacher for revisions. Describe in detail all procedures and experimental design including methods for data collection, and when applicable, the source of data used. When writing out the steps, please list them out numerically. Do not write this as a paragraph. **Describe only your project. Do not include work done by your mentor or others.** If you will use published surveys, questionnaires, or tests, describe how you obtained these, including required permission if applicable. This should be super detailed. Meaning,*

that an outside person, when reading this list, could perform the experiment if the supplies were available.)

1. Prepare Orange Pi:
2. Flash Orange Pi OS (Arch or Droid) onto a 32GB+ microSD card using BalenaEtcher.
3. Insert the card and mount the Orange Pi 5B inside the project box.
4. Attach PCA9685:
5. Solder header pins onto the PCA9685 16-Channel Servo Driver.
6. Safety warning: Wear eye protection and work in a ventilated area.
7. Connect PCA9685:
8. Connect PCA9685 to the Orange Pi's I2C pins using jumper wires (SCL, SDA, VCC, GND).
9. Attach Camera Module:
10. Gently insert the ribbon cable from the Raspberry Pi Camera Module into the Orange Pi's CSI port.
11. Wire and Mount MPU-6050:
12. Connect the MPU-6050 to the Orange Pi's I2C pins using jumper wires.
13. Mount the module securely on the car chassis, away from motors.
14. Mount and Connect LiDAR:
15. Mount the Yahboom Silan LiDAR at the highest point on the car.
16. Connect the LiDAR to the Orange Pi's serial port.
17. Power the System:
18. Use a portable USB power bank to power both the PCA9685 (servos) and the Orange Pi.
19. Wire Motors and Servos:
20. Connect the RC car's motors (via ESCs) and steering servos to the correct channels on the PCA9685.
21. Secure Electronics:
22. Use hot glue to mount the project box to the car chassis, securing all wires.
23. Safety during operation: Supervise the car during autonomous testing.
24. Set up MongoDB:
25. Install MongoDB on a separate server and configure it for remote access.
26. Connect Remotely:
27. Use SSH to connect to the Orange Pi over Wi-Fi.
28. Test Components:
29. Use Python libraries (smbus, pyserial) to test communication with all sensors and the servo driver.
30. Install Reinforcement Learning (RL) Framework:
31. Install Stable-Baselines3 and create a custom Gymnasium environment in Python.
32. Define the observation space (position, heading, map) and action space (steering, throttle).
33. Create reward functions for exploration, collision avoidance, and navigation.
34. Define Exploration Environment:
35. Use a large, obstacle-rich indoor space for testing.
36. Randomize the starting position and orientation for each trial.

37. Measure Exploration:
38. Use LiDAR to generate a 2D map.
39. Define "exploration" as achieving 95% map coverage.
40. End trials when 95% coverage is reached or a time limit (e.g., 30 min) expires.
41. Conduct 20 Trials:
42. Run 20 fully autonomous trials, each starting from a random position.
43. Log all relevant data (trial number, time, position, LiDAR, IMU, actions, rewards) to MongoDB.
44. Compile and Analyze Data:
45. Retrieve data from MongoDB.
46. Use a Python plotting library like Matplotlib to create a graph visualizing performance metrics (e.g., time to 95% coverage over 20 trials).

Risk and Safety: *(Identify any potential risks and safety precautions needed. Every experiment has risk and safety concerns, even very minor and unlikely possibilities. This should not be left blank.)*

- Flammability, sparks, and heat burns → Wear fire-protective gear (protective gloves, fire extinguisher/blanket)
- Getting cut, Shrapnel hazards → Adult supervision and eye/hand-protection gear
- Electrical hazards, shocks → Use electric-insulating gloves

Data Analysis: *(Describe how you will analyze or compare your data to determine your results. (Will you use tables, graphs or formulas?))*

Dependent variables: speed, coverage

Speed is measured by the stopwatch when the rovers are running.

Coverage is measured by the GPS updates inside each rover.

All the GPS processing data will be transferred into a matrix form and stored in a cloud database.

$X = \text{speed}$, $Y = \text{coverage}$, $Y/X = \text{efficiency}$

An average score will be taken after the 20 trials.

Bibliography:

Type in your Bibliography: *(Use <https://www.mybib.com/> to properly cite your sources. Why do you need this section? You should be researching to get the background necessary to develop these ideas and plans. List as many sources as needed.)*

NASA. "Mars Pathfinder." NASA Mars Internal Resources, mars.nasa.gov/internal_resources/815/. Accessed 13 Oct. 2025.

Possible Extra Information:

Choose the appropriate sections needed for your project. Then fill out those sections on the following pages. **Delete the sections you do not need.**

No extra sections needed.